

$$\begin{aligned}
3.1 \quad (1) \quad P\{Y(t+\tau) - Y(t) = n\} &= P\{X_1(t+\tau) + X_2(t+\tau) - X_1(t) - X_2(t) = n\} \\
&= P\{X_1(t+\tau) - X_1(t) + X_2(t+\tau) - X_2(t) = n\} \\
&= \sum_{i=0}^n P\{X_1(t+\tau) - X_1(t) = i, X_2(t+\tau) - X_2(t) = n-i\} \\
&= \sum_{i=0}^n P\{X_2(t+\tau) - X_2(t) = n-i\} \cdot P\{X_1(t+\tau) - X_1(t) = i\} \\
&= \sum_{i=0}^n e^{-\lambda_1 \tau} \cdot \frac{(\lambda_1 \tau)^i}{i!} \cdot e^{-\lambda_2 \tau} \cdot \frac{(\lambda_2 \tau)^{n-i}}{(n-i)!} = e^{-\lambda \tau} \cdot \frac{(\lambda \tau)^n}{n!}
\end{aligned}$$

$$\lambda = \lambda_1 + \lambda_2$$

$$(2) \quad E Z(t) = E[X_1(t) - X_2(t)] = E X_1(t) - E X_2(t) = (\lambda_1 - \lambda_2)t.$$

$$D Z(t) = D[X_1(t) - X_2(t)] = D X_1(t) + D X_2(t) = (\lambda_1 + \lambda_2)t$$

$\therefore E Z(t) \neq D Z(t)$, $\therefore Z(t)$ 不是泊松过程.

3.2

顾客数 $\{N(t), t \geq 0\}$ 服从泊松分布, 令 ξ_i 表示第 i 个顾客购物与否,

$$\xi_i = \begin{cases} 1, & \text{第 } i \text{ 个顾客购物} \\ 0, & \text{第 } i \text{ 个不购物} \end{cases}$$

$$P(\xi_i = 1) = p, \quad P(\xi_i = 0) = 1 - p$$

$$Y(t) = \sum_{i=1}^{N(t)} \xi_i \text{ 是复合泊松过程, } E Y(t) = \lambda p t.$$

$$\text{强度 } \lambda_Y = \lambda p.$$

3.3

$$(1) \quad P(X(2) = 3) = \frac{(2\lambda)^3}{3!} e^{-2\lambda} = \frac{8\lambda^3}{6} e^{-2\lambda} = \frac{4}{3} \lambda^3 \cdot e^{-2\lambda}$$

$$\begin{aligned}
(2) \quad P &= \sum_{k=0}^2 P\{X(1) - X(0) = k, X(2) - X(1) \geq 3 - k\} \\
&= \sum_{k=0}^2 P\{X(1) - X(0) = k\} P\{X(12) - X(1) \geq 3 - k\}
\end{aligned}$$

$$\begin{aligned}
&= \frac{\lambda^0}{0!} e^{-\lambda} \left(1 - \frac{\lambda^0}{0!} e^{-\lambda} - \frac{\lambda^1}{1!} e^{-\lambda} - \frac{\lambda^2}{2} e^{-\lambda} \right) + \lambda e^{-\lambda} \left(1 - \frac{\lambda^0}{0!} e^{-\lambda} - \lambda e^{-\lambda} \right) \\
&\quad + \frac{\lambda^2}{2} e^{-\lambda} \left(1 - \frac{\lambda^0}{0!} e^{-\lambda} \right) \\
&= e^{-\lambda} \left(1 - e^{-\lambda} - \lambda e^{-\lambda} - \frac{\lambda^2}{2} e^{-\lambda} \right) + \lambda e^{-\lambda} (1 - e^{-\lambda} - \lambda e^{-\lambda}) \\
&\quad + \frac{\lambda^2}{2} e^{-\lambda} (1 - e^{-\lambda}) \\
&= \underbrace{e^{-\lambda}} - \underbrace{e^{-2\lambda}} - \underbrace{\lambda e^{-2\lambda}} - \frac{\lambda^2}{2} e^{-2\lambda} + \underbrace{\lambda e^{-\lambda}} - \underbrace{\lambda e^{-2\lambda}} - \underbrace{\lambda^2 e^{-2\lambda}} + \underbrace{\frac{\lambda^2}{2} e^{-\lambda}} - \frac{\lambda^2}{2} e^{-2\lambda} \\
&= e^{-\lambda} \left(1 + \lambda + \frac{\lambda^2}{2} \right) - e^{-2\lambda} (1 + 2\lambda + 2\lambda^2)
\end{aligned}$$

3.4.

$$P\{S > s_1 + s_2 \mid S > s_1\} = P\{X(s_1 + s_2) - X(s_1) = 0\} = \frac{(\lambda s_2)^0}{0!} e^{-\lambda s_2} = e^{-\lambda s_2}$$

$$\text{从而 } P\{S > s_2\} = 1 - P\{S \leq s_2\} = 1 - (1 - e^{-\lambda s_2}) = P\{S > s_1 + s_2 \mid S > s_1\}$$

3.5

$$(1) f_{\text{爆}}(t) = \begin{cases} \lambda_1 e^{-\lambda_1 t}, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

(2) 由 3.1 可知, 多个独立的泊松过程可合成一个新泊松过程 $Y(t)$

$$f_Y(t) = \begin{cases} (\lambda_1 + \lambda_2 + \lambda_3) e^{-(\lambda_1 + \lambda_2 + \lambda_3)t}, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

3.6.

$$\begin{aligned}
P(W_n \leq t) &= \sum_{k=n}^{\infty} \frac{(\lambda t)^k}{k!} e^{-\lambda t} \\
f_{W_n}(t) &= \sum_{k=n}^{\infty} \left[\frac{(\lambda t)^{k-1}}{(k-1)!} e^{-\lambda t} - \lambda \frac{(\lambda t)^k}{k!} e^{-\lambda t} \right] \\
&= \sum_{k=n}^{\infty} \left[\frac{(\lambda t)^{k-1}}{(k-1)!} e^{-\lambda t} \left(1 - \frac{\lambda t}{k} \right) \right]
\end{aligned}$$

设 T_i 表示 $\{X(t), t \geq 0\}$ 第 $i-1$ 次事件发生到第 i 次事件发生的时间间隔,
 则 $T_i (i=1, 2, \dots, n)$ 相互独立且都服从均值为 $\frac{1}{\lambda}$ 的指数分布.

$$ET_i = \frac{1}{\lambda}, \quad DT_i = \frac{1}{\lambda^2}, \quad i=1, 2, \dots, n$$

$$(1) \quad EW_n = E\left[\sum_{i=1}^n T_i\right] = \sum_{i=1}^n ET_i = \frac{n}{\lambda}$$

$$(2) \quad DW_n = D\left[\sum_{i=1}^n T_i\right] = \frac{n}{\lambda^2}$$

★ 到达时刻分布的结论.

3-7. 构造条件概率

$$P\{N=k\} = \int_0^\infty P\{Y(w') - Y(w) = k, w' - w = s\} ds$$

$$= \int_0^\infty P\{Y(w') - Y(w) = k \mid w' - w = s\} P\{w' - w = s\} ds$$

$$= \int_0^\infty \frac{(\lambda_2 s)^k}{k!} \cdot e^{-\lambda_2 s} \cdot \lambda_1 \cdot e^{-\lambda_1 s} ds$$

$$= \frac{\lambda_2^k}{k!} \lambda_1 \int_0^\infty s^k \cdot e^{-(\lambda_1 + \lambda_2)s} ds \dots \quad (1)$$

$$\text{其中, } \int_0^\infty s^k \cdot e^{-(\lambda_1 + \lambda_2)s} ds = -\frac{1}{\lambda_1 + \lambda_2} \int_0^\infty s^k d e^{-(\lambda_1 + \lambda_2)s}$$

$$= -\frac{1}{\lambda_1 + \lambda_2} (0 - k \int_0^\infty s^{k-1} \cdot e^{-(\lambda_1 + \lambda_2)s} ds)$$

$$= \frac{1}{\lambda_1 + \lambda_2} \cdot k \int_0^\infty s^{k-1} \cdot e^{-(\lambda_1 + \lambda_2)s} ds$$

$$\text{令 } G_k = \int_0^\infty s^k \cdot e^{-(\lambda_1 + \lambda_2)s} ds$$

$$\text{则 } G_k = \frac{1}{\lambda_1 + \lambda_2} \cdot k \cdot G_{k-1}$$

$$= \frac{1}{(\lambda_1 + \lambda_2)^2} \cdot k \cdot (k-1) \cdot G_{k-2}$$

$$\dots$$

$$= \frac{1}{(\lambda_1 + \lambda_2)^k} \cdot k! \cdot G_0$$

$$\text{又 } G_0 = \int_0^\infty e^{-(\lambda_1 + \lambda_2)s} ds = \frac{1}{\lambda_1 + \lambda_2}$$

$$\begin{aligned} \therefore \textcircled{1} &\Rightarrow \frac{\lambda_2^k}{k!} \lambda_1 \cdot \frac{1}{(\lambda_1 + \lambda_2)^k} \cdot k! \cdot \frac{1}{\lambda_1 + \lambda_2} \\ &= \frac{\lambda_1}{\lambda_1 + \lambda_2} \cdot \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^k \end{aligned}$$

3.8

与3.2类似, $EX(t) = EN(t) \cdot E\delta_1 = \lambda p t$, $X(t)$ 是强度为 λp 的复合泊松过程.

$$\therefore P\{X(t) = k\} = e^{-\lambda p t} \frac{(\lambda p t)^k}{k!}$$